builds and deployments for dynamic requirements.

- ***Community and ecosystem:*** With an active community, Jenkins X benefits from contributed plugins, extensions, and best practices, ensuring long-term success as a CI/CD tool.
- ***Flexibility:*** Highly configurable and extensible, JX allows organisations to tailor and integrate it with existing toolsets, providing flexibility in implementation.

Advantages of Jenkins X with respect to other CI/CD tools

Jenkins vs Jenkins X

Concept

Jenkins: Monolithic CI/CD tool with extensive plugin support for automation.

Jenkins X: Cloud-native CI/CD solution designed for Kubernetes, emphasising microservices, GitOps, and automation.

Working

Jenkins: Manual setup for various configurations, integration, and pipeline automation.

Jenkins X: Native Kubernetes support, built-in GitOps, and automated CI/CD pipelines, reducing manual intervention and promoting best practices for modern development in Kubernetes environments.

GitHub Actions vs Jenkins X

Concept

GitHub Actions: Integrated CI/CD within GitHub for automated workflows directly tied to repositories.

Jenkins X: Cloud-native CI/CD solution designed for Kubernetes, emphasising microservices, GitOps, and automation, with a focus on scalability and multi-environment support.

Working

GitHub Actions: Native GitHub repository integration, YAML-based workflows, and built-in CI/CD capabilities.

Jenkins X: Seamlessly integrates with GitHub repositories, providing native Kubernetes support, built-in GitOps, and automated CI/CD pipelines. Tailored for cloud-native development in Kubernetes environments, it offers inherent scalability and supports multi-environment promotion, providing a comprehensive solution beyond GitHub Actions.

Cloud-based CI/CD and Jenkins X

In the context of continuous integration/continuous deployment (CI/CD), 'cloud-based CI/CD' and 'Jenkins X' are not directly comparable because they have various functions and reflect different paradigms, but we can provide a brief synopsis of each and point out a few differences.

Concept

Cloud-based CI/CD:

- Deployment on cloud platforms with scalable, managed CI/CD services.
- Emphasises pre-configured environments for streamlined setup.

Jenkins X:

- Cloud-native CI/CD solution designed for Kubernetes, suitable for cloud or on-premises deployment.
- Focuses on microservices, GitOps, and automation, providing flexibility and control.

Working

Cloud-based CI/CD:

- Hosted on the cloud, it offers pre-configured environments for CI/CD processes.
- Integrates with cloud services and resources seamlessly.
- Maintenance and updates are outsourced for high availability.

Jenkins X:

- Cloud-native, supports Kubernetes, and provides built-in GitOps and automation.
- Tailored for Kubernetes environments with multi-environment support.
- Requires hands-on management and customisation for flexibility and control in the CI/CD process.

Installation procedure

The installation prerequisites for Jenkins X can vary depending on your environment, but here are some general steps and prerequisites for setting it up.

Kubernetes cluster*:* Ensure you have a functional Kubernetes cluster. This could be a local setup (e.g., Minikube) or a cloud-based solution like GKE, EKS, or AKS.

kubectl*:* Install kubectl, the command-line tool used to interact with Kubernetes clusters.

Helm*:* Jenkins X utilises Helm for managing Kubernetes applications. Install Helm to deploy Jenkins X and manage its components.

Git*:* You'll need Git installed on your machine to interact with version control repositories.

GitOps*:* GitOps is essential for Jenkins X to ensure declarative management of infrastructure and applications, enabling version-controlled, auditable, and automated deployments through Git repositories.

Docker*:* Jenkins X works with containers, so having Docker installed can be helpful for local testing and development.

Access credentials*:* If deploying to a cloud provider, have appropriate access credentials or API tokens for that provider.

jx CLI*:* Install the jx command-line tool, which is specific to Jenkins X. This tool helps with managing Jenkins X installations, applications, and pipelines.

Once you've ensured these prerequisites are met, the installation process usually involves running a set